

- 2 An algorithm has been written in pseudocode to generate 50 positive random integers with values less than or equal to 100. These random integers are stored in the array `RandNum[ ]`

The function `Rand(X, Y)` generates a random integer greater than or equal to `X` and less than `Y`. For example, `Rand(1, 4)` generates 1 or 2 or 3.

```
1 Count ← 0
2 REPEAT
3     RandNum[Counter] ← Rand(1, 100)
4     Count ← Count + 2
5 UNTIL Count <= 50
```

- (a) Find the **four** errors in the pseudocode and write a correction for each error.

Error 1

Correction

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Error 2

Correction

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Error 3

Correction

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Error 4

Correction

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[4]

- (b) The pseudocode for this algorithm could be shortened by the use of a `FOR ... NEXT` loop.

Rewrite the algorithm using a `FOR ... NEXT` loop.

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..... [3]

- (c) Identify another loop structure available in pseudocode.

..... [1]

4 The pseudocode algorithm should work as a calculator and output the result.

```
1  Continue ← 1
2  WHILE Continue = 0
3      OUTPUT "Enter 1 for +, 2 for -, 3 for * or 4 for /"
4      INPUT Operator
5      OUTPUT "Enter the first value"
6      INPUT Value1
7      OUTPUT "Enter the second value"
8      OUTPUT Value2
9      IF Operator
10         1: Answer ← Value1 + Value2
11         2: Answer ← Value1 - Value2
12         3: Answer ← Value1 * Value2
13         4: Answer ← Value1 / Value2
14     ENDCASE
15     OUTPUT "The answer is ", Value1
16     OUTPUT "Do you wish to enter more values (Yes or No)?"
17     INPUT MoreValues
18     IF MoreValues = "No"
19         THEN
20             Continue ← 1
21     ENDIF
22 UNTIL Continue = 0
```

(a) Find the **five** errors in the pseudocode and suggest a correction for each error.

Error 1

Correction

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Error 2

Correction

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Error 3

Correction

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Error 4

Correction

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Error 5

Correction

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- (b)** The algorithm needs changing to allow only the numbers 1, 2, 3, or 4 to be entered for the input variable `Operator`.

Write the pseudocode to perform this task and state where in the algorithm it would be located.

Pseudocode

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Location in algorithm

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- 2 An algorithm has been written in pseudocode to generate 50 positive random integers with values less than or equal to 100. These numbers are stored in the array NumRand[]

The function RandUp (X, Y) generates a random integer greater than X and less than or equal to Y
For example, RandUp (1, 4) generates 2 or 3 or 4

```
1 Count ← 0
2 WHILE Counter > 50 DO
3     NumRand[Counter] ← RandUp (1,100)
4     Counter ← Counter - 2
5 ENDWHILE
```

- (a) Find the **four** errors in the pseudocode and write a correction for each error.

Error 1

Correction

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Error 2

Correction

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Error 3

Correction

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Error 4

Correction

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[4]

- (b) The pseudocode for this algorithm could be shortened by the use of a FOR ... NEXT loop.

Rewrite the algorithm using a FOR ... NEXT loop.

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[3]

- input the ages of 100 students
- count and output the number of students aged 7 and under 12
- count and output the number of students aged 12 and under 18
- count and output the number of students aged 18 and over.

(a) Complete the pseudocode algorithm:

```

01 Count7to12 ← 0
02 Count12to18 ← 0
03 CountOver18 ← 0
04 FOR Student ← 1 TO .....
05     OUTPUT "Please enter student's age in years "
06     INPUT Age
07     IF Age >= 7 .....
08         THEN
09             Count7to12 ← Count7to12 + 1
10     ENDIF
11     IF Age >= 12 AND Age < 18
12         THEN
13             Count12to18 ← .....
14     ENDIF
15     IF Age >= 18
16         THEN
17             CountOver18 ← CountOver18 + 1
18     ENDIF
19 NEXT Student
20 OUTPUT "There are ", Count7to12, " students aged 7 and under 12."
21 OUTPUT "There are ", Count12to18, " students aged 12 and under 18."
22 OUTPUT "There are ", ....., " students aged 18 and over."

```

[4]

(b) Write the extra pseudocode statements that are needed to count and output the number of students under the age of 7. Use the variable `CountUnder7`; assume `CountUnder7` has already been set to zero.

[4]

- [4]

[4]

- 2 An algorithm allows a user to input their password and checks that there are at least eight characters in the password. Then, the user is asked to re-input the password to check that both inputs are the same. The user is allowed three attempts at inputting a password of the correct length and a matching pair of passwords. The pre-defined function `LEN(X)` returns the number of characters in the string, `X`

```
01 Attempt ← 0
02 REPEAT
03   PassCheck ← TRUE
04   OUTPUT "Please enter your password "
05   INPUT Password
06   IF LEN(Password) < 8
07     THEN
08       PassCheck ← TRUE
09     ELSE
10       OUTPUT "Please re-enter your password "
11       INPUT Password2
12       IF Password <> Password2
13         THEN
14           PassCheck ← FALSE
15         ENDIF
16       ENDIF
17       Attempt ← Attempt + 1
18 UNTIL PassCheck OR Attempt <> 3
19 IF PassCheck
20   THEN
21     OUTPUT "Password success"
22   ELSE
23     OUTPUT "Password fail"
24   ENDIF
```

- (a) Identify the **three** errors in the pseudocode and suggest a correction to remove each error.

Error 1

Correction

Error 2

Correction

Error 3

Correction

(b) The algorithm includes **two** types of check on the data input.
Identify and describe each type of check.

Type of check 1

Description

.....

Type of check 2

Description

.....

[4]

(c) Give **two** sets of test data for this algorithm and a reason for choosing each set.

Each set of test data and its reason must be different.

Set 1

Reason

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Set 2

Reason

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[4]

3 (a) Describe a one-dimensional array. Include an example of an array declaration.

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[3]

(b) Explain how indexing could be used to search for a value stored in a one-dimensional array.

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[2]